

Why wild norm residue symbols are hard

Tame symbols live in the residue field; wild ones live in the p -adic logarithm. The depth, the group sizes and the failure of every shortcut, computed

on a question of Rene Pannekoek; everything checked in wild-symbols.gp, output in results/wild-symbols.txt

The one-line answer. For $v \nmid n$ the symbol is a function on the residue field: units pair trivially, $K^\times/(K^\times)^n$ has n^2 elements, and the value is one Legendre-type exponentiation in $\mathbb{F}_q$. For $v \mid n$ none of that survives: units pair non-trivially, the group has $n^2/|n|_K$ elements, and the symbol depends on its arguments to depth $pe/(p-1)$ in the maximal ideal. **Tame is algebraic, wild is analytic** — the honest formulas involve p -adic logarithms and formal power series, not residues. You already know one wild symbol: the quadratic symbol at 2, the one that needs a and b **modulo 8** rather than modulo p .

1. The two regimes

For a local field K with residue field $\mathbb{F}_q$ containing μ_n , the n -th norm residue symbol is

$$(\cdot, \cdot)_n : K^\times/(K^\times)^n \times K^\times/(K^\times)^n \longrightarrow \mu_n, \quad (a, b)_n = \left(\sigma_a \left(\sqrt[n]{b} \right) \right) / \sqrt[n]{b},$$

σ_a being the local Artin symbol. It is 1 exactly when a is a norm from $K \left(\sqrt[n]{b} \right)$.

Tame, $v \nmid n$. The extension $K \left(\sqrt[n]{b} \right) / K$ is tamely ramified, and the Artin map on units factors through the residue field. The symbol is given in closed form by

$$(a, b)_n \equiv (-1)^{v(a)v(b)(q-1)/n} \left(a^{v(b)} / b^{v(a)} \right)^{(q-1)/n} \pmod{\mathfrak{m}},$$

one exponentiation in $\mathbb{F}_q^\times$. For $n=2$ and odd p this is the familiar $(a, b)_p = ((-1/p))^{\alpha\beta} ((u/p))^\beta ((w/p))^\alpha$ with $a = p^\alpha u$, $b = p^\beta w$ — checked here against PARI on 195792 cases, 0 disagreements.

Everything below is what breaks when $v \mid n$.

2. Break 1: units stop being invisible

If $v \nmid n$ then $U^{(1)} = 1 + \mathfrak{m}$ is a pro- p group and n is prime to p , so $U^{(1)}$ is **uniquely n -divisible**: it sits inside $(K^\times)^n$ and dies in the quotient. The symbol cannot see units at all.

Over all odd $p < 200$ and all pairs of units, 556656 symbols were computed and **every one is trivial**. At $v=2$ the four unit classes pair like this:

$(u, w)_2$	1	3	5	7
1	+1	+1	+1	+1
3	+1	-1	+1	-1
5	+1	+1	+1	+1
7	+1	-1	+1	-1

$(3, 3)_2 = -1$: two principal units pairing non-trivially. At an odd place that is impossible whatever the units are. Wildness **is** the statement that the unit filtration survives into $K^\times/(K^\times)^n$, and the symbol therefore has a filtration to see.

2.1. The pairing has depth

Which layers pair? At $v = 2$ the symbol is non-trivial on $U^{(i)} \times U^{(j)}$ exactly when $i + j \leq 2$:

(i, j)	$(1, 1)$	$(1, 2)$	$(2, 2)$	$(i + j \geq 3)$
non-trivial?	yes	no	no	no

That bound $i + j \leq pe/(p - 1)$ is the general shape. The tame symbol has no filtration at all: it is trivial on $U \times U$ outright.

3. Break 2: how much of a and b the symbol sees

At an odd p , changing a or b by p never changes the symbol — checked for all odd $p < 100$, 0 violations. **The tame symbol reads its arguments modulo p .**

At $v = 2$ it does not. 1 and 5 agree modulo 4, but

$$(1, 2)_2 = +1, \quad (5, 2)_2 = -1.$$

Modulo 4 is not enough; modulo 8 is (checked on 10^4 odd pairs, 0 violations). This is the wildness everyone has met without naming it.

The depth, in general. Write $e = v_K(p)$. Then

$$U^{(m)} \subseteq (K^\times)^n \quad \text{exactly when} \quad m > (pe)/(p - 1).$$

For $\mathbb{Q}_2$ and $n = 2$: $e = 1$, $pe/(p - 1) = 2$, so $m \geq 3$ — modulo 8. Verified.

3.1. The same computation for cubes

The interesting case is an odd p , where nobody has memorised the answer. Take $K = \mathbb{Q}_3(\zeta_3)$, $\mathcal{O} = \mathbb{Z}[\omega]$ with $\omega^2 = -1 - \omega$, $\pi = \omega - 1$ and $\pi^2 = -3\omega$, so $e = v_\pi(3) = 2$ and $pe/(p - 1) = 3$. The prediction is $m \geq 4$. Brute force over $\mathcal{O}/\pi^{12} = \mathcal{O}/3^6$ (well past the Hensel threshold $2e + 1 = 5$, so “cube mod π^{12} ” really means “cube”):

m	units $\equiv 1 \pmod{\pi^m}$	all cubes?
0	354294	no (341172 fail)
1	177147	no (170586 fail)
2	59049	no (52488 fail)
3	19683	no (13122 fail)
4	6561	yes
5	2187	yes
6	729	yes

Exactly $m \geq 4 = pe/(p - 1) + 1$, on the nose. Note the last failing row: at $m = 3$ precisely one third of the units are cubes — there is exactly one layer of the filtration left, and the symbol is what detects it.

This is the same constant as hensel-different.typ. That document asks why square classes need a modulo 8 at $v = 2$ and only a modulo v at odd v , and answers: the intrinsic invariant is the Kähler different, $e = v(\det J)$, and Hensel’s threshold is $2e + 1$ — so 3 at 2 and 1 elsewhere. Here the same e controls the same phenomenon from the other side: $2e + 1$ is the (non-sharp) sufficient bound, $pe/(p - 1)$ the sharp one. For $\mathbb{Q}_2$ they agree at 3; for cubes at 3 Hensel gives 5 where the truth is 4. **Wild ramification is a statement about the different, and precision is what it costs.**

4. Break 3: the group explodes

$$|K^\times/(K^\times)^n| = n^2/|n|_K.$$

Tame means $|n|_K = 1$ and the answer is n^2 . Wild means $|n|_K < 1$, and the factor $|n|_K^{-1}$ is the whole story. For $n = 2$: four classes over odd $\mathbb{Q}_p$, eight over $\mathbb{Q}_2$ — the familiar $\{\pm 1, \pm 2, \pm 5, \pm 10\}$.

For $n = p$ the symbol is only defined over $K = \mathbb{Q}_p(\zeta_p)$, which is totally ramified of degree $p - 1$, so $e = p - 1$, $|p|_K^{-1} = p^{p-1}$ and $|K^\times/(K^\times)^p| = p^{p+1}$:

p	tame answer p^2	wild answer p^{p+1}	depth $pe/(p-1) = p$
3	9	81	3
5	25	15625	5
7	49	5764801	7
11	121	3138428376721	11
13	169	3937376385699289	13

For $p = 11$ — the case that came up in `ec-density-bm.typ` — any formula for the symbol has to be a function on a group with $3.1 \cdot 10^{12}$ elements, not 121.

5. Break 4: acquiring μ_n makes it worse

The symbol is not even **defined** until $\mu_n \subseteq K$. For $n = p$ odd this forces $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$, totally ramified of degree $p - 1$ — so the fix for one problem deepens the other:

field	$e = v(p)$	depth $pe/(p-1)$	$[K : \mathbb{Q}_p]$
$\mathbb{Q}_p$ (no μ_p)	1	< 2	1
$\mathbb{Q}_3(\zeta_3)$	2	3	2
$\mathbb{Q}_5(\zeta_5)$	4	5	4
$\mathbb{Q}_7(\zeta_7)$	6	7	6
$\mathbb{Q}_{11}(\zeta_{11})$	10	11	10

For $n = 2$ this costs nothing: $\mu_2 = \{\pm 1\}$ is already in $\mathbb{Q}_2$, $e = 1$, depth 2, and the whole answer fits in a mod-8 formula. **That is why the quadratic symbol at 2 is the one wild symbol with a closed form** — and the only one in PARI or Sage.

6. What the explicit formulas actually look like

The wild symbol is not uncomputable; it is that no residue-field shortcut exists, so any formula must encode a p -adic analytic object.

- **Artin–Hasse (1928)**, for $K = \mathbb{Q}_p(\zeta_{p^n})$ and u a principal unit, express the symbol in the shape

$$(u, \cdot)_{p^n} = \zeta_{p^n}^{\text{Tr}_{K/\mathbb{Q}_p}(\dots \log u)/p^n},$$

a p -adic logarithm followed by a trace. Computable — but you must carry $\log u$ to precision $pe/(p-1)$ in a field of degree $p-1$, and the formulas cover only special second arguments (ζ , or the uniformiser).

- **General explicit reciprocity laws** — Iwasawa, Wiles, Coleman, Vostokov, Brückner and others — handle general local fields and general n . They are power-series formulas: Coleman series, residues of differential forms, formal-group logarithms. Each is a theorem with real content, not a symbol you evaluate in a line.
- **Lubin–Tate** theory makes the local Artin map completely explicit via formal groups, so “computable in principle” is fair. But it is a formal-group computation, not a finite-field one.

So the honest summary of “hard”. Not undecidable, not even deep — just: no shortcut through the residue field, hence p -adic analysis to depth $pe/(p-1)$, in a field you had to ramify by $p-1$ to get into, on a class group of size p^{p+1} . Every one of those four numbers is 1 or trivial in the tame case.

7. Lubin–Tate: where the constant comes from

Lubin–Tate theory builds the maximal totally ramified abelian extension of K out of a formal group F_π over $\mathcal{O}_K$ with

$$[\pi](x) = \pi x + \dots + x^q,$$

and makes the local Artin map completely explicit: a uniformiser acts trivially on $K(F_\pi[\pi^n])$ and by Frobenius on $K^\times$, while a unit u acts on $F_\pi[\pi^n]$ by $[u^{-1}]$. So the norm residue symbol is a formal-group computation, and “computable in principle” is exactly what that buys.

For $K = \mathbb{Q}_p$ and $\pi = p$ the Lubin–Tate group is the multiplicative one, $\hat{\mathbb{G}}_m$, and $K_\pi = \mathbb{Q}_p(\mu_{p^\infty})$ — so Lubin–Tate recovers the cyclotomic picture and the Artin–Hasse formulas of Section 6 rather than replacing them. Wiles’ explicit reciprocity law is the Lubin–Tate generalisation of Artin–Hasse, and Coleman power series are what make it work.

7.1. The depth is a formal-group statement

What the formal group explains outright is the constant $pe/(p-1)$ that ran through Section 3. On $\hat{\mathbb{G}}_m$,

$$[p](x) = (1+x)^p - 1 = px + \binom{p}{2}x^2 + \dots + x^p,$$

so with $j = v(x)$ and $e = v_K(p)$ the two extreme terms compete:

$$v(px) = e + j, \quad v(x^p) = pj.$$

	$v(px)$	$v(x^p)$	which wins
$j < e/(p-1)$	$e + j$	pj	x^p — Frobenius regime , valuation multiplied by p
$j = e/(p-1)$	$e + j$	pj	tie — the crossover
$j > e/(p-1)$	$e + j$	pj	px — linear regime , valuation shifted by e

Above the crossover $[p]$ carries $\mathfrak{m}^j$ onto $\mathfrak{m}^{j+e}$, so its image contains $\mathfrak{m}^m$ as soon as $m \geq e/(p-1) + e$, which is the threshold $m > pe/(p-1)$. Evaluated:

K	e	$e/(p-1)$	$pe/(p-1)$	$U^{(m)}$ are p -th powers for
$\mathbb{Q}_2$	1	1	2	$m \geq 3$
$\mathbb{Q}_3(\zeta_3)$	2	1	3	$m \geq 4$
$\mathbb{Q}_5(\zeta_5)$	4	1	5	$m \geq 6$
$\mathbb{Q}_{11}(\zeta_{11})$	10	1	11	$m \geq 12$

The first two rows are exactly the thresholds measured by brute force in Section 3 — mod 8 for squares in $\mathbb{Q}_2$, and $m \geq 4$ for cubes in $\mathbb{Q}_3(\zeta_3)$.

So “wild” has a one-line formal-group meaning. Below $j = e/(p-1)$ the map $[p]$ behaves like **Frobenius**, $x \mapsto x^p$, multiplying valuations by p ; above it like **multiplication by p** , shifting valuations by e . The crossover is precisely where $\log_F$ converges and becomes an isomorphism $F(\mathfrak{m}^j) \cong \mathfrak{m}^j$ — which is why the symbol turns analytic there, and why every explicit reciprocity law is a formula in logarithms.

In the tame case there is no crossover to cross. n is prime to p , so $[n]$ is invertible on the formal group and the whole unit filtration is n -divisible; the formal group never enters the computation, and one is left with the residue field. **Tame means the Lubin–Tate group is invisible.**

8. Where this bit these notes

- `descent-s3.typ` needed a **cubic** norm-residue symbol at the wild place $v = 3$ to exhibit a Brauer class, and recorded that neither PARI nor Sage provides one. Its README entry notes the gap is “an implementation task at one wild place, not a missing tool”.
- `ec-density-bm.typ` §6 hits the same wall one prime up: the index-11 failure of density for the conductor-89 curve at $p = 11$ needs an order-11 Brauer class, and the symbol that has to be non-trivial is the 11-th norm residue symbol **at 11**. Depth 11, degree 10, $3.1 \cdot 10^{12}$ classes.
- And the structural reason it is always the wild symbol there: $E_1(\mathbb{Q}_p) \cong \mathbb{Z}_p$ is pro- p , so cutting the formal-group layer always demands a class of order p evaluated at the place p . The tame layers of the Néron filtration are the reachable ones.

Companion file: `wild-symbols.gp`, run as

```
gp -q -s 120000000000 wild-symbols.gp < /dev/null > results/wild-symbols.txt
```

It verifies the triviality of the tame symbol on units (556656 cases), the mod- p versus mod-8 precision, the depth of the unit filtration for squares in $\mathbb{Q}_2$ and cubes in $\mathbb{Q}_3(\zeta_3)$ (brute force over $\mathcal{O}/3^6$, 531441 elements), the pairing filtration at 2, the class-group sizes, and the tame closed formula against PARI’s `hilbert` on 195792 cases.